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Glass chemistry, paleomagnetism, and correlation of middle Pleistocene tuffs in southern North Island, New Zealand, and Western Pacific

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Abstract New Zealand middle Pleistocene (Castlecliffian) rhyolitic tuffs older than 0.5 Ma are characterised on the basis of glass chemistry and paleomagnetism for the purpose of correlation. Of the tuffs examined, Rewa Pumice (c. 0.74 Ma), Potaka Pumice (c. 0.64 Ma), and Lower Te Muna Tephra (c. 0.75 Ma) are particularly widespread marker horizons. Their proven proximity to the Brunhes–Matuyama paleomagnetic reversal provides a useful framework for correlation and absolute age control between regions in southern North Island—Wanganui, Wairarapa, and Hawke's Bay—and with deep-sea cores from the Western Pacific. The tuffs allow correlation between a diverse range of sedimentary facies including marine, freshwater, and terrestrial, and provide absolute age control for thick sequences of unfossiliferous strata in Wairarapa and Hawke's Bay.

The characterisation and correlation of the tuffs has a number of implications for Castlecliffian paleoclimatic, tectonic, and stratigraphic studies. The Rewa Pumice is contained in periglacial facies, and we assign it to oxygen isotope stage 20. Some tuffs in Hawke's Bay contain charcoal and are considered to be derived from ignimbrites. Their presence implies that the main axial ranges in Hawke's Bay were uplifted after 0.6 Ma. Previous lithostratigraphic correlations from the Castlecliffian stratotype along the Wanganui coast to sections across the Wanganui Basin, which contain radiometrically dated tuffs, are not sustainable, suggesting greater facies variations than hitherto assumed. The sequence of strata in the Rangitikei section containing the Rewa and Potaka Pumices is not a correlative of Kaimatira Pumice Sand at the coast section. Previous tuff correlations were made on the basis of age and mineralogy. Only the integrated use of glass chemistry and paleomagnetism allows correlation of tuffs in this time interval.

Keywords tephrostratigraphy; magnetostratigraphy; tuffs; correlation; Pleistocene; Castlecliffian; Wanganui; Wairarapa; Hawke's Bay; Western Pacific; glass chemistry

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INTRODUCTION

Thick sequences of middle Pleistocene (Castlecliffian Stage) strata occur in several sedimentary basins of southern North Island, New Zealand, and contain rhyolitic, mainly waterlaid, tephra (tuffs). Prominent tuff horizons in a section exposed by the Rangitikei River in the Wanganui region (Fig. 1) have been radiometrically dated and provide absolute age control for the Castlecliffian Stage (c. 0.4–1.0 Ma; Seward 1974). Prominent tuff horizons also occur in Castlecliffian strata of southern Hawke's Bay and Wairarapa (Fig. 1). Tuff correlations have been proposed on stratigraphic and lithological grounds (Fleming 1953; Lillie 1953; Collen & Vella 1984) but have not been substantiated by any unique characteristics of the tuffs, such as glass chemistry. Correlation between the regions is further hindered by the predominance of alluvial and lacustrine facies in southern Hawke's Bay and Wairarapa, which do not allow biostratigraphic correlation to mainly marine strata exposed in the Wanganui area.

Numerous deep-sea cores from abyssal sediments in the Western Pacific, east of New Zealand, contain tuffs older than 0.5 Ma. Correlations to onshore sequences have been proposed based on tuff ages from magnetostratigraphically derived sedimentation rates and a few fission-track dates (Watkins & Huang 1977; Kyle & Seward 1984; Seward et al. 1986). These correlations have not been tested by glass chemistry. Relating the New Zealand Castlecliffian Stage to oxygen isotope stages (e.g., Beu & Edwards 1984; Kamp & Turner 1990) requires the precise location of magnetic reversal boundaries in sections and the recognition of tuff beds for age control. The latter requires chemical characterisation to avoid miscorrelation otherwise possible on the basis of lithology and stratigraphic position alone.

We have chemically characterised glass shards by electron microprobe from Castlecliffian tuffs in each region, as well as from two deep-sea cores, for the purpose of correlation. We attempt to extend the late Pleistocene tephrostratigraphic framework of Froggatt (1983) to strata older than 0.5 Ma. Paleomagnetic reversals occurring before 0.5 Ma (Brunhes–Matuyama and Jaramillo Event) provide an opportunity to use reversal boundaries as a framework for tuff correlations. Therefore, paleomagnetic samples from each region were obtained.

Three principal onshore sections were selected for investigation, which consist of a thick and relatively continuous Castlecliffian sequence: Rangitikei River, Wanganui (400 m); Huangarua River, Wairarapa (366 m; Collen & Vella 1984); and Mangatewaiiti Stream, southern Hawke's Bay (c. 600 m, new data). Other, shorter sequences were also examined (Fig. 1; Appendix 1).

GLASS CHEMISTRY

Preparation of tuff samples for glass chemistry is described by Froggatt & Gosson (1982). Glass shards were analysed by a JEOL 733 Superprobe at the Analytical Facility of Victoria

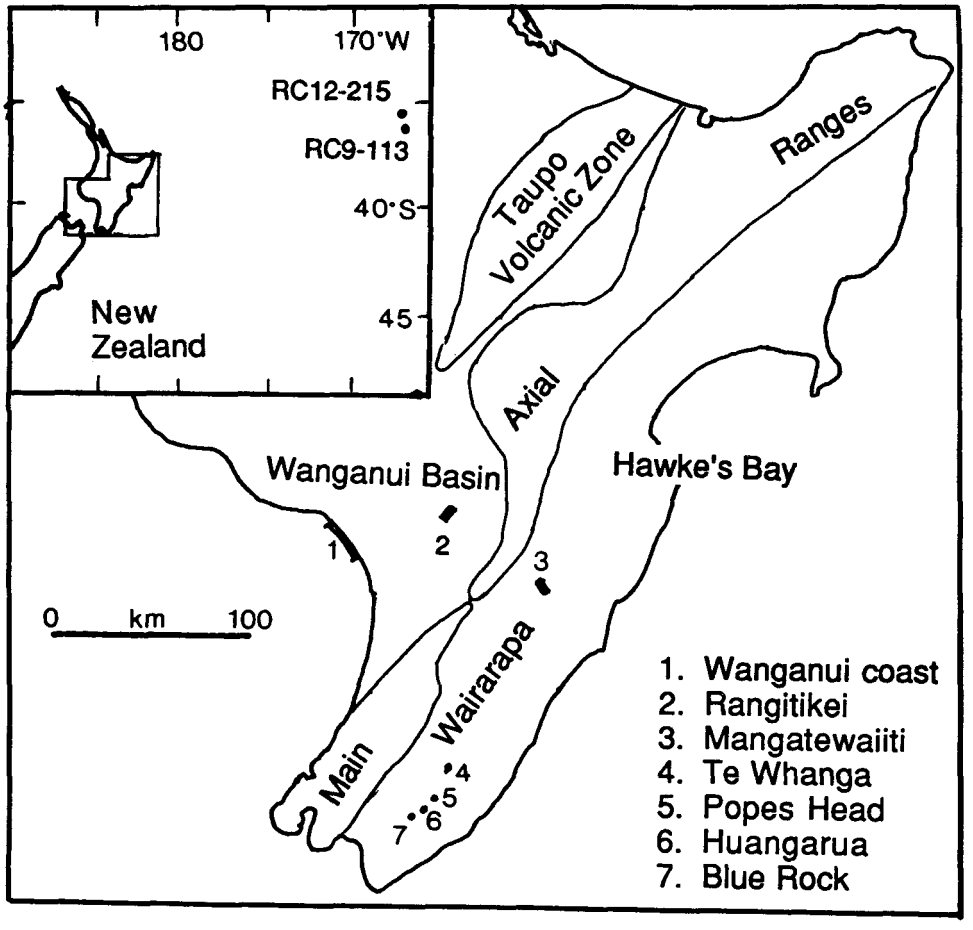


Fig. 1 Locality of sections and cores examined in this study.

University of Wellington, using an 8 nA beam at 15 kV and 10 μm diameter and glass standards as in Froggatt (1983).

Electron microprobe analyses of glass shows all the tuffs are calc-alkaline rhyolites (SiO_2 75.17–78.52 wt%: Table 1). The compositions are broadly similar to those of late Quaternary tephra and lavas (Cole 1979; Froggatt 1983) of the Taupo Volcanic Zone. The Taupo Volcanic Zone is considered the source of the tuffs since voluminous rhyolitic volcanic rocks of comparable age to Rangitikei tuffs are known only from that region (e.g., Cole 1979).

Most tuffs are chemically homogeneous, consisting of a single glass population presumably associated with a single eruption or a series of closely spaced eruptions. Homogeneity is shown by the low standard deviation in analyses for most oxides, such as FeO (< 0.15) and TiO_2 and MgO (< 0.04) (Froggatt 1983). Two tuffs: S020 (Te Whanga) and S075/S100 (Mangatawaiiti Stream) consist of multiple glass populations and have been grouped into separate populations by inspection, where possible (Table 1). Multiple glass populations can be easily distinguished by variation diagrams (Fig. 2). Such mixed populations probably result from erosion and sedimentary reworking of different eruptive products (Froggatt 1983).

PALEOMAGNETISM

Paleomagnetic core samples were obtained from the Rangitikei, Huangarua, and Mangatewaiiti sections and tuff exposures at Popes Head and Te Whanga in Wairarapa (Table 2). Remanence of the cores was measured on a Molspin Spinner Magnetometer and they were demagnetised by standard alternating field (AF) methods. Natural Remanent Magnetism (NRM) intensities varied from c. 65.0 to 0.13 mA/m.

Samples were selected from each stratigraphic section for stepwise AF cleaning in increments of 5 mT. Examples of good results are shown in Fig. 3. Others were cleaned in fewer steps dependent on the behaviour of the pilot samples. Most samples had median destructive field strengths (50% of initial field remaining) at c. 20 mT. Samples not displaying orderly magnetic intensity decay and high consistency in field orientation within sites were discarded.

Samples from many sites from the Rangitikei section, in particular those within 30 m above or below the Rewa Pumice, behaved erratically on cleaning (Fig. 4). Similar results for this interval were obtained by Seward (1974) who used thermal cleaning methods. We found many samples changed from a normal to a reversed declination during demagnetisation, but inclinations remained normal (negative), suggesting an incomplete removal of a strong normal overprint.

STRATIGRAPHY AND CORRELATIONS

A summary of tephra correlations and magnetostratigraphy along with previous paleomagnetic data is shown in Fig. 5. Tuffs are correlated by glass chemistry (Table 3) using similarity coefficients (SC) (Froggatt 1983; Sarna-Wojcicki et al. 1987). SC values greater than 0.92, using all elements from microprobe analyses except Cl, indicate close similarity and are nearly always obtained for correlative tuffs. A correlation is not possible unless the pair of tuffs occurs in strata of similar age,

Table 1 Microprobe analyses of glass shards, recalculated to 100%. Means and standard deviations on *n* analyses, – = not determined. Water by difference, all Fe as FeO. See Froggatt (1983) for standard analyses. Operating conditions were 10 μm beam, 8 nA at 15 kV. Letters following analysis number refer to subpopulations within the sample.

	1	2	3	4	5	6	7	8
SiO_2	78.32(.45)	75.70(1.4)	76.21(.82)	78.21(.66)	77.17(.37)	77.67(.29)	76.42(.56)	77.24(.35)
Al_2O_3	12.40(.15)	13.11(.37)	13.08(.28)	12.34(.66)	12.73(.21)	12.23(.16)	13.21(.21)	12.51(.12)
TiO_2	0.09(.02)	0.14(.04)	0.16(.03)	0.13(.04)	0.13(.04)	0.11(.01)	0.17(.03)	0.13(.03)
FeO	1.26(.07)	2.10(.14)	2.02(.20)	1.13(.10)	1.65(.07)	1.15(.16)	1.76(.32)	1.11(.08)
MgO	0.10(.02)	0.11(.03)	0.14(.04)	0.11(.02)	0.12(.02)	0.11(.03)	0.15(.04)	0.11(.02)
CaO	1.16(.09)	0.98(.13)	1.05(.16)	0.86(.16)	0.77(.06)	0.94(.16)	1.08(.05)	0.93(.09)
Na_2O	3.42(.25)	4.29(.27)	3.98(.27)	3.53(.19)	4.00(.20)	3.55(.19)	3.82(.21)	4.07(.12)
K_2O	3.20(.36)	3.16(.36)	3.24(.47)	3.54(.47)	3.33(.25)	4.05(.31)	3.39(.14)	3.67(.22)
Cl	0.17(.05)	0.21(.03)	0.19(.02)	0.22(.03)	–	0.23(.04)	–	0.23(.01)
Water	5.72(.79)	5.74(.64)	5.93(1.94)	5.28(1.04)	–	6.09(2.28)	4.98(.89)	5.52(.47)
n	27	78	24	34	12	10	9	9
	9	10	11	12	13	14A	14b	15A
SiO_2	75.27(.20)	75.42(.33)	75.74(.57)	76.81(.68)	76.68(.28)	76.13(.37)	77.50(.35)	76.53(.38)
Al_2O_3	13.56(.09)	13.51(.23)	13.59(.13)	13.04(.11)	13.23(.22)	13.02(.14)	77.50(.35)	76.50(.38)
TiO_2	0.17(.02)	0.17(.02)	0.14(.03)	0.16(.05)	0.16(.03)	0.16(.04)	0.10(.03)	0.17(.04)
FeO	2.07(.09)	2.15(.08)	2.03(.09)	2.02(.06)	1.92(.17)	1.93(.11)	1.38(.07)	2.03(.10)
MgO	0.12(.03)	0.12(.01)	0.12(.02)	0.13(.03)	0.10(.02)	0.09(.02)	0.07(.03)	0.12(.03)
CaO	1.06(.04)	1.05(.06)	1.05(.06)	1.10(.10)	1.00(.04)	1.00(.08)	0.80(.08)	1.07(.08)
Na_2O	4.16(.29)	4.03(.38)	3.94(.55)	3.83(.30)	3.54(.34)	4.01(.13)	3.75(.14)	3.93(.14)
K_2O	3.59(.36)	3.56(.20)	3.39(.21)	3.26(.08)	3.37(.13)	3.48(.14)	3.87(.22)	3.40(.14)
Cl	–	–	–	0.20(.03)	–	0.18(.02)	0.17(.02)	0.18(.03)
Water	4.92(.88)	5.23(1.00)	5.53(.82)	5.25(1.09)	–	3.92(1.06)	3.74(.57)	5.16(88)
n	9	9	10	11	18	14	6	22
	15B	15C	15D	16	17	18	19	20
SiO_2	78.16(.38)	75.17(.51)	78.51(.23)	76.70(.23)	78.14(.28)	78.26(36)	78.16(.23)	77.77(.47)
Al_2O_3	12.08(.21)	13.11(.23)	11.61(.08)	12.45(.12)	11.99(.14)	12.44(.16)	12.49(.19)	12.54(.14)
TiO_2	0.09(.04)	0.25(.04)	0.09(.04)	0.18(.03)	0.10(.02)	0.08(.02)	0.10(.03)	0.12(.04)
FeO	1.33(.08)	2.48(.06)	1.43(.07)	2.01(.11)	1.04(.12)	0.93(.09)	0.95(.18)	1.13(.11)
MgO	0.07(.03)	0.19(.04)	0.03(.02)	0.11(.03)	0.09(.03)	0.11(.02)	0.13(.08)	0.11(.02)
CaO	0.93(.03)	1.37(.05)	0.67(.03)	1.06(.05)	0.86(.12)	0.85(.10)	0.83(.11)	0.86(.11)
Na_2O	3.35(.18)	3.97(.38)	3.64(.10)	4.11(.22)	3.44(.13)	3.70(.23)	3.64(.16)	3.94(.41)
K_2O	3.82(.19)	3.30(.13)	3.80(.06)	3.21(.14)	4.12(.15)	3.63(.32)	3.70(.23)	3.53(.25)
Cl	0.17(.05)	0.18(.07)	0.22(.03)	0.17(.05)	0.22(.03)	–	–	–
Water	4.69(.66)	3.79(2.13)	6.46(2.70)	6.03(1.25)	3.77(2.02)	3.64(.92)	3.99(1.48)	4.27(1.57)
n	10	4	4	17	11	16	18	19
	21	22	23	24	25			
SiO_2	78.11(.37)	75.62(.23)	78.33(.25)	78.15(.32)	77.76(.28)			
Al_2O_3	12.32(.15)	13.44(.22)	12.38(.17)	12.39(.20)	12.49(.18)			
TiO_2	0.13(.02)	0.16(.03)	0.09(.03)	0.12(.04)	0.13(0.3)			
FeO	1.02(.11)	1.89(.12)	0.90(.13)	1.02(.11)	1.16(.11)			
MgO	0.11(.02)	0.15(.02)	0.10(.02)	0.12(.02)	0.12(.02)			
CaO	0.86(.04)	1.30(.12)	0.82(.12)	0.79(.05)	0.87(.05)			
Na_2O	3.57(.34)	4.39(.13)	3.73(.14)	3.95(.11)	4.05(.15)			
K_2O	3.88(.18)	3.06(.15)	3.65(.28)	3.46(.16)	3.42(.16)			
Cl	–	–	–	–	–			
Water	6.19(.65)	6.00(.72)	5.50(.80)	4.95(1.12)	4.70(.82)			
n	19	17	18	19	17			

Sample identities are:

Wanganui Basin tuffs:

1 Pakihikura Pumice
2 Mangapipi Ash
3 Rewa Pumice
4 Potaka Pumice
5 Kaukatea Ash

Wairarapa and Hawke's Bay tuffs:

6 Kaimatira Pumice Sand
7 Upper Te Muna
8 Lower Te Muna
9 Upper Popes Head
10 Lower Popes Head
11 Akupe Tephra
12 Blue Rock Tuff
13 Te Whanga Tuff
14 S020
15 S100
16 S075
17 S099

Deep-sea core samples:

RC9-113

18 A
19 B
20 C

RC12-215

21 M3
22 M4
23 A
24 B
25 C

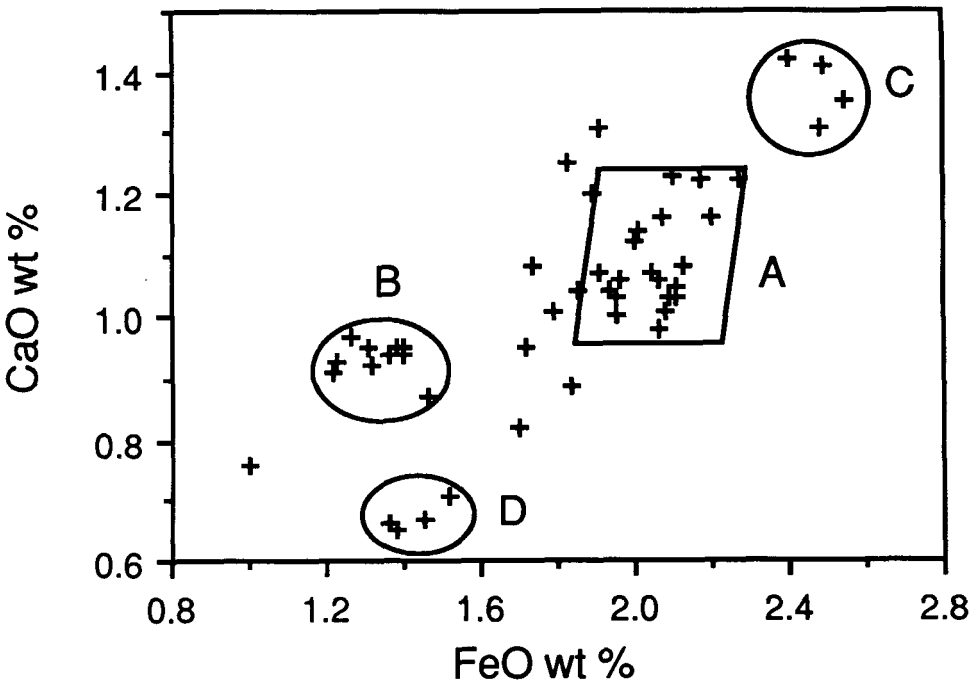
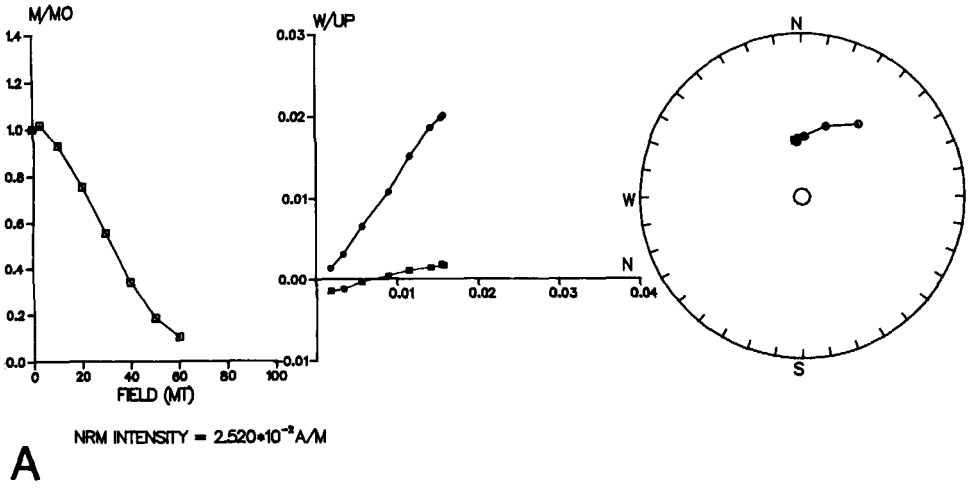


Fig. 2 Multiple glass populations as shown by individual shard analyses of tuff S100 from Mangatewaiiti Stream. Populations A, B, C, and D are defined on the basis of all oxides analysed for, not just FeO and CaO shown here. Population A is chemically identical to Rewa Pumice (see text).



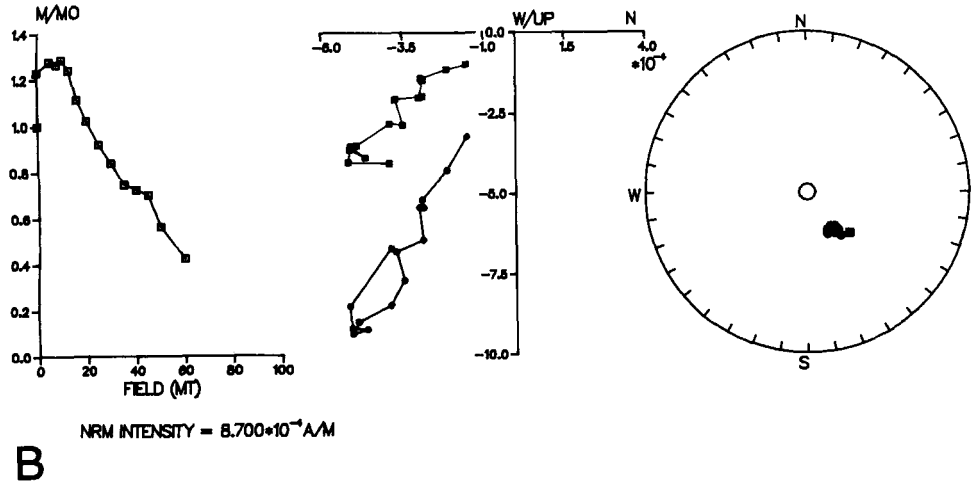


Fig. 3 Examples of good results of stepwise demagnetisation, shown on modified Zijderveld diagrams and stereographic projections of the resulting fields. On Zijderveld plots squares and dots refer to x versus y and x versus z components, respectively. Graph at left shows the ratio of intensity to initial intensity of NRM during demagnetisation. A, Normal polarity sample from Huangarua River mudstone. B, Reversed polarity sample from Te Whanga tuff.

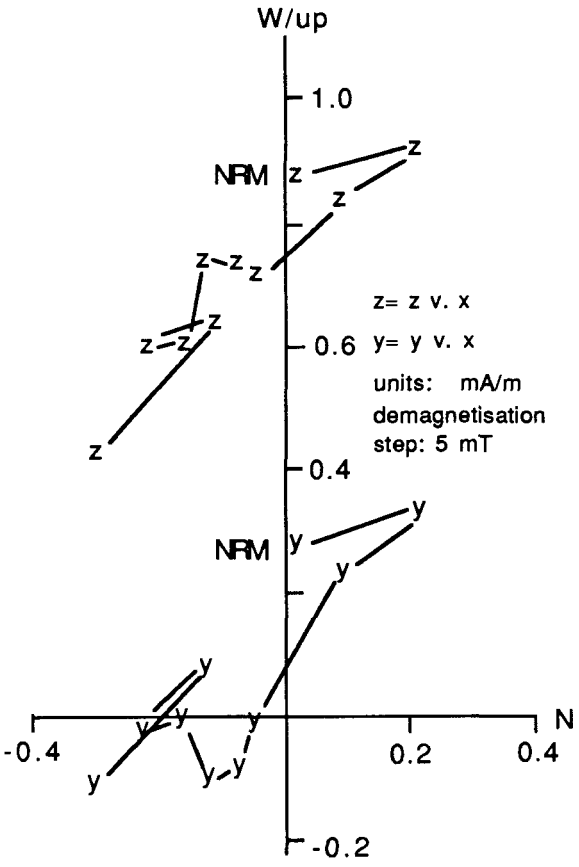


Fig. 4 An example of the behaviour of specimens from the Rewa Pumice interval in the Rangitikei section during demagnetisation, plotted on a Zijderveld diagram. This specimen is from 20 m below the Rewa Pumice. No stable end point is reached; however, demagnetisation suggests a reversed polarity with a strong normal overprint.

stratigraphic position, and the same magnetic polarity, where such data exist.

Rangitikei section

The stratigraphy of the Rangitikei section has been previously described by Te Punga (1952) and Seward (1974). Seward (1974, 1979) dated four prominent tuff horizons from the section: Pakihikura Pumice (1.06 ± 0.16, 1.06 ± 0.18 Ma, fission-track on glass and zircon, respectively); Mangapipi Ash (0.88 ± 0.13 Ma, glass); Rewa Pumice (0.74 ± 0.09 Ma, glass); and Potaka Pumice (0.61 ± 0.06, 0.64 ± 0.18 Ma, glass and zircon). The Rewa Pumice and Potaka Pumice are also known as the lower and upper Kaimatira Pumice Sand, respectively, but this correlation is discussed below. In addition, we examined two other tuffs from the Wanganui area: Kaukatea Ash (0.57 ± 0.08 Ma, glass) and Kaimatira Pumice Sand (undated) from the Wanganui coastal section.

Seward (1974) tentatively located the Brunhes–Matuyama boundary between the Rewa Pumice and Potaka Pumice in the Rangitikei section. We, also, could not obtain reliable paleomagnetic data from the Rewa Pumice interval (Fig. 4, 5), however, reversed declinations found during demagnetisation suggest reversed polarity. Some samples from above the tuff did display reversed polarity, but with wide scatter within sites. Our data from the Potaka Pumice confirm that this tuff is of normal polarity and the Pakihikura Pumice and Mangapipi Ash are of reversed polarity (Table 2).

Each of the above tuffs at Rangitikei are multilayered and display evidence of extensive reworking in an intertidal and/or estuarine environment. Multiple samples of each are chemically homogeneous, so we assumed that each represents multiple depositional events from a single eruption. These analyses are grouped together in Table 1. Each of the Rangitikei section tuffs and the Kaukatea Ash can be distinguished on the basis of their glass chemistry (Table 3). The Mangapipi Ash and Rewa Pumice are most similar (SC = 0.93); however, the former is distinguished by its higher Na_2O and lower TiO_2 and MgO contents (Table 1).

Huangarua section and southern Wairarapa

The thickest known sequence of Castlecliffian strata in southern Wairarapa is exposed in the Huangarua River (366 m) and is thought to represent most of the Castlecliffian Stage (Collen & Vella 1984). Only two tuff horizons occur in the section. We examined eight tuffs from four localities in southern Wairarapa (see Appendix 1), all of which occur in a similar stratigraphic position. Electron microprobe analyses (Table 1) of the eight tuff beds allows them to be divided into three chemically distinct groups: (a) Upper and Lower Popes Head Tephras, Akupe Tephra, Upper Te Muna Tephra, Blue Rock Tuff, and Te Whanga Tuff; (b) Lower Te Muna Tephra; and (c) sample S020. Group (a) tuffs occur stratigraphically above group (b), since the Upper Te Muna Tephra occurs 0.5 m above the Lower Te Muna Tephra in Huangarua River (Fig. 5). At the Popes Head section, three tuffs occur: Upper and Lower Popes Head Tephras and Akupe Tephra (Fig. 5). The former two merge along strike indicating reworking, and because all three tuffs are stratigraphically close and contained in channels, we consider all to represent a single eruption. Group (c) tuff consists of multiple glass populations.

Group (a) tuffs occur in a diverse range of sedimentary facies. At Huangarua, the tuff occurs in loess; and in Blue Rock Stream, the Blue Rock Tuff overlies a weathered paleosol. At Popes Head on Hinakura Road, the three reworked tuff beds are contained in small channels interpreted to be ox-bow lake deposits. At Te Whanga, the correlative is current bedded and contained within a fluvial sequence.

Table 2 Paleomagnetic polarity of some Pleistocene tuffs, with α_{95} values calculated on n samples, after cleaning at 20 mT. Samples with positive inclinations have reversed polarity.

Site	Dec.	Inc.	n	α_{95}
Rangitikei				
Potaka Pumice (top)	345	–63	2	9
Potaka Pumice (base)	337	–62	5	13
Mangapipi Ash	192	40	3	20
Pakihikura Pumice	175	29	3	17
Mangatewaiiti				
S099 pumice	061	–53	3	21
S100 pumice (top)	006	–53	4	8
S100 (middle)	075	–31	3	7
S100 (base)	178	32	4	10
Huangarua				
Lower Te Muna	183	66	3	9
Popes Head				
Upper Popes Head	174	60	3	24
Akupe Tephra	175	57	3	18
Te Whanga				
Te Whanga Tuff	161	63	3	14

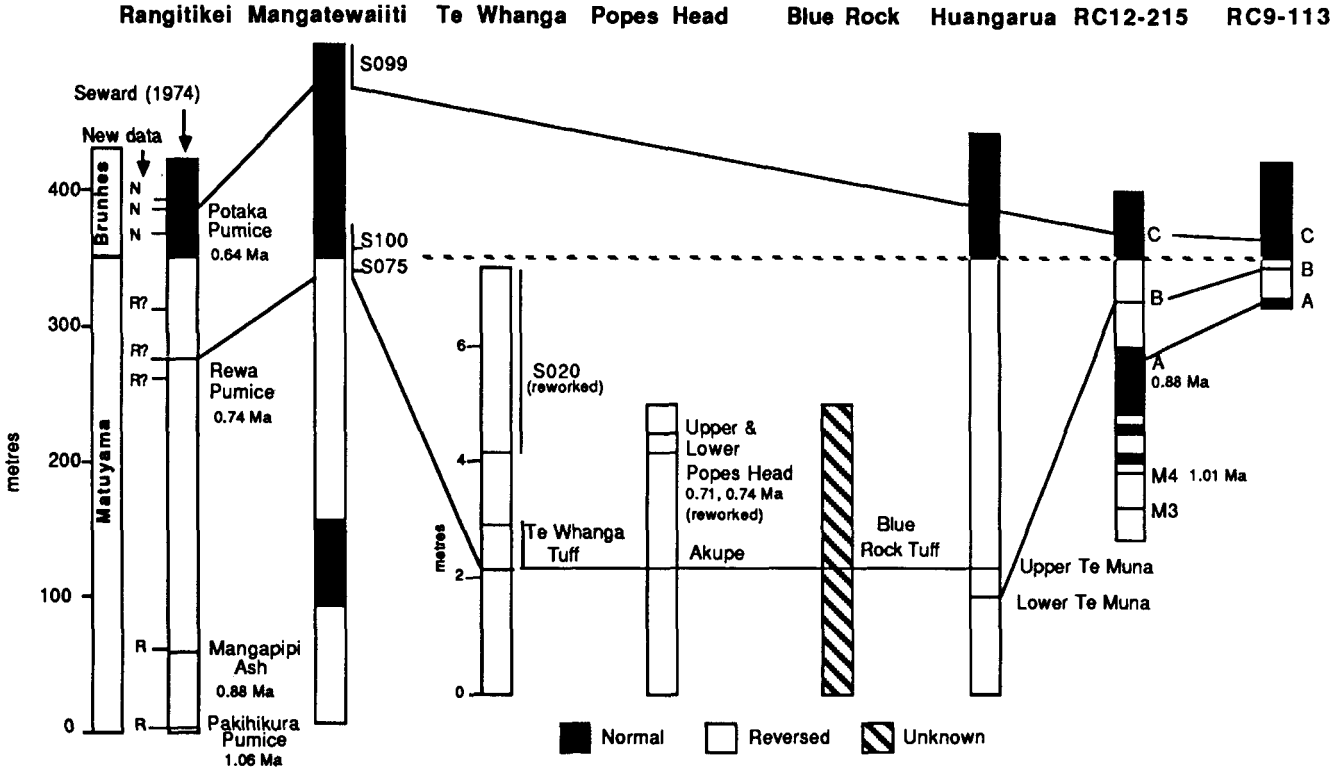


Fig. 5 Summary of Castlecliffian tuff correlations established by glass chemistry and magnetostratigraphy. Also shown are fission-track dates (Seward 1974, 1979). Magnetostratigraphy for cores RC12–215 and RC9–113 are from Watkins & Huang (1977). The Mangatewaiiti data are from 15 sample sites for this part of the section and are a summary of ongoing work. Note scale change for the Te Whanga section, which also applies to all sections to the right of it.

Collen & Vella (1984) have suggested the Brunhes–Matuyama boundary occurs near the middle of Castlecliffian strata exposed in Huangarua River (type section for Te Muna Formation). However, the polarity of that part of the section which contains the Te Muna Tephras was then unknown. The cleaned paleomagnetic data we obtained (Table 2; Fig. 5) show the tuffs are reversed and the boundary occurs c. 5 m above them. Our data from the Upper Popes Head Tephra, Akupe Tephra, and Te Whanga Tuff show they are of reversed polarity (Table 2). This is consistent with estimated fission-track ages of 0.71 ± 0.14 and 0.74 ± 0.16 Ma (annealed glass) from the Upper and Lower Popes Head Tephras, respectively, by Seward (1979).

Tuffs of group (a) are chemically identical to the Rewa Pumice (SC range 0.93–0.98) and dissimilar to other Rangitikei tuffs (Table 3). They show some similarity to the Mangapipi Ash; however, the Rewa Pumice provides a better match in having higher TiO_2 and SiO_2 content and lower Na_2O content. The correlation to Rewa Pumice (0.74 Ma) is also supported by the close stratigraphic position of the Brunhes–Matuyama boundary (c. 0.74 Ma) to the Wairarapa tuffs and the estimated fission-track ages for the Popes Head Tephras.

The Lower Te Muna Tephra is chemically similar to the Potaka Pumice (SC = 0.96; Table 1); however, they are not correlatives as they have disparate magnetic polarities (Table 2).

Mangatewaiiti section

Castlecliffian strata in southern Hawke's Bay is represented by the Mangatarata Formation (Lillie 1953). A thick sequence (c. 600 m) is exposed in Mangatewaiiti Stream north of Dannevirke (Fig. 1). The upper c. 200 m of the section contains two fluvially deposited tuffaceous units (S099 and S075/100) interbedded with lacustrine mudstone. These tuffs are coarse grained (lapilli) and contain charcoal which suggests an ignimbritic origin. The upper tuff, S099, is located within a normal magnetic polarity interval, while the lower tuff, S075/100, displays normal polarity at its top and reversed polarity at its base (Fig. 5; Table 2). The lower tuff is a multiple depositional unit and contains considerable hiatuses consistent with it representing an extended period of deposition. Tuffs S099 and S075/100 can be chemically matched to the Potaka and Rewa Pumices, respectively (Table 3), which is consistent with the upper reversal boundary in the section being the Brunhes–Matuyama, as the Jaramillo Event has been located lower in the section (Fig. 5). One sample (S075 in Table 1) from the lower tuff is relatively chemically homogeneous, whereas sample S100 from a higher stratigraphic level is not and contains a population ('A' in Fig. 2) which is identified as Rewa Pumice (Table 3).

Deep-sea cores, Western Pacific

We have analysed five prominent megascopic tuffs of middle Pleistocene age (0.6–1.2 Ma) in deep-sea abyssal sediments contained in piston cores RC9–113 and RC12–215, from the Western Pacific, east of New Zealand (Fig. 1). They have ages determined from sedimentation rates from magnetostratigraphy and a few fission-track ages (Watkins & Huang 1977; Seward 1974; Kyle & Seward 1984): M3 (1.2 Ma), M4 (1.07 Ma), A (0.88 Ma), B (0.75 Ma), and C (0.69 Ma). Tuff C has previously been correlated to onshore exposures in New Zealand using

glass chemistry and magnetostratigraphy (Froggatt 1983). It is considered to be the Potaka Pumice.

We correlate the Lower Te Muna Tephra exposed in Wairarapa to tuff B in RC12–215 and RC9–113 (Fig. 5, Table 3). Tuff B (0.75 Ma) occurs in reversed magnetic polarity sediments in both cores, a short distance below the Brunhes–Matuyama boundary. The Lower Te Muna Tephra occurs a short distance below the Upper Te Muna Tephra (= Rewa Pumice, 0.74 Ma) and in a similar magnetostratigraphic position. The recognition of correlatives of both the Lower Te Muna Tephra and Potaka Pumice in the same core (RC12–215, Fig. 5) confirms the existence of two chemically identical tuffs (Table 3) of different magnetic polarity and different stratigraphic position.

DISCUSSION

Implications for Castlecliffian stratigraphy

We have demonstrated tuff correlations over distances of 200–300 km between Castlecliffian strata of different regions of southern North Island and over 1000 km with deep-sea cores (Fig. 5). The Rewa Pumice (0.74 Ma), Potaka Pumice (0.64 Ma), and the Brunhes–Matuyama paleomagnetic boundary provide absolute age control and correlation between the radiometrically dated tuffs in the Rangitikei section, Mangatarata Formation in southern Hawke's Bay, and Te Muna Formation in Wairarapa. This confirms the age of unfossiliferous alluvial and lacustrine strata proposed for the latter two regions (Lillie 1953; Collen & Vella 1984). Correlation of Potaka Pumice and Lower Te Muna Tephra to tuffs in Western Pacific deep-sea cores on which magnetostratigraphy are available further support the identification of the Brunhes–Matuyama boundary in Hawke's Bay and Wairarapa.

Seward (1976) has proposed the correlation (on the basis of mineralogy and trace elements in titanomagnetites) of the Potaka Pumice in the Rangitikei section to the type section of Kaimatira Pumice Sand at Kaimatira Bluff and to pumice in the nearby Castlecliffian stratotype exposed along the Wanganui coast (Fleming 1953). The glass chemistry of shards in the Wanganui coast section is very similar to that of the Potaka Pumice (Tables 1, 3). However, Kamp & Turner (1990) have identified the Brunhes–Matuyama boundary 30 m above pumice designated as Kaimatira Pumice Sand in the coastal section. This makes correlation to Potaka Pumice unlikely on the basis of the magnetostratigraphy we have presented (Fig. 5). Instead, we suggest the pumice at Kaimatira Bluff and Wanganui coast correlates with the Lower Te Muna Tephra (= layer B in deep-sea cores) which occurs below the Brunhes–Matuyama boundary in reversed polarity strata and has identical chemistry to Potaka Pumice (Fig. 5; Tables 1, 3). Therefore, no simple correlation between radiometrically dated tuffs in the Rangitikei section and the Castlecliffian stratotype can yet be made. The usefulness of the Kaimatira Pumice Sand as a lithostratigraphic unit for correlation is greatly limited since it appears to represent different tuffs in different areas.

Previous workers have attempted to correlate Castlecliffian tuffs exposed in the Rangitikei section to tuffs in deep-sea cores (layers M3, M4, A, B and C) on the basis of ages from radiometric methods and sedimentation rates (Watkins & Huang 1977; Kyle & Seward 1984; Seward et al. 1986). None of these correlations are possible on the basis of glass chemistry and paleomagnetism, except correlation of Potaka Pumice to layer C (Froggatt 1983). We now correlate layer B to the Lower Te Muna Tephra using these methods. This demonstrates

Table 3 *Similarity coefficients (see text for details) comparing the glass chemistry between the tuffs analysed and the major recognised tuffs in Wanganui Basin and the Lower Te Muna Tephra. Highest values indicate close chemical similarity. Italicised values are correlations consistent with stratigraphic position and magnetostratigraphy.*

	Pakihikura	Mangapipi	Rewa	Potaka	Kaukatea	Lower Te Muna
Pakihikura	1	84	82	89	83	87
Mangapipi		1	93	87	91	90
Rewa			1	84	77	85
Potaka				1	91	96
Kaukatea					1	92
Lo Te Muna						1
Kaimatira	90	85	82	95	87	*94*
Up. Te Muna	82	89	*95*	84	88	85
Up. Popes	81	93	*95*	84	89	86
Lo. Popes	81	93	*95*	84	89	86
Akupe	83	95	*95*	87	92	88
Blue Rock	83	93	*98*	85	89	85
Te Whanga	86	93	*93*	88	89	87
S020A	83	93	*93*	84	89	88
S100A	83	93	*96*	86	91	87
S075	84	94	*95*	86	89	88
S099	89	79	77	*91*	83	89
A(RC9-113)	88	82	79	92	85	91
B(RC9-113)	87	81	82	92	87	*91*
C(RC9-113)	88	88	84	*97*	92	97
M3	87	86	82	97	89	95
M4	80	89	94	81	85	82
A(RC12-215)	90	81	78	91	85	89
B(RC12-215)	86	85	84	94	93	*94*
C(RC12-215)	87	88	87	*96*	82	96

the inadequacy of using tuff ages alone for correlation and the need for an approach integrating chemistry, paleomagnetism, and chronology. The use of the ages of the Rangitikei tuffs to assign stratigraphic units elsewhere to oxygen isotope stages (e.g., Wanganui coast; Beu & Edwards 1984) is not supported by our correlations.

Paleoclimatic and tectonic implications

The stratigraphic position of the Upper Te Muna Tephra (= Rewa Pumice) as an airfall deposit within loess in the Huangarua section indicates the eruptive event occurred during a glacial interval shortly before the Brunhes–Matuyama magnetic reversal (Fig. 5). At Mangatewaiiti Stream, a thick unit of coarse alluvial greywacke conglomerate occurs within reworked Rewa Pumice at the Brunhes–Matuyama boundary. Alluvial conglomerate is also associated with Rewa Pumice at Te Whanga. Collen & Vella (1984) suggested conglomerate in Castlecliffian strata of Wairarapa represents alluvial aggradation during glacial intervals of increased erosion and low sea level. Therefore, on the basis of magnetostratigraphy and association with periglacial facies, we assign the Rewa Pumice to glacial oxygen isotope stage 20 (Shackleton & Opdyke 1976).

The occurrence of thick conglomerate beds within reworked Rewa Pumice at Mangatewaiiti Stream represents the first significant appearance of coarse nonvolcanic alluvium in the section (i.e., at 0.74 Ma). This could represent the initiation of uplift of the main axial ranges of North Island in that region (Ruahine – southern Hawke's Bay), consistent with previous estimates (Beu et al. 1981). Alternatively, the production of the conglomerate is climatically controlled, as Rewa Pumice has been assigned to glacial stage 20, and the beginning of rapid uplift is younger. Thick (>20 m), fluvially deposited, coarse tuffs containing charcoal indicative of derivation from ignimbrites, occur above this interval at Mangatewaiiti Stream. They indicate the axial ranges were not a barrier to fluvial transport and ignimbrite flow from the Taupo Volcanic Zone, until well after the deposition of Potaka Pumice (0.64 Ma).

CONCLUSIONS

Of the middle Pleistocene (Castlecliffian) tuffs examined in southern North Island and in deep-sea cores of Western Pacific, the Rewa Pumice, Potaka Pumice, and Lower Te Muna Tephra (= layer B) are particularly widespread marker horizons. They occur as thick deposits at great distances from their source (Taupo Volcanic Zone) and, except for Rewa Pumice, as megascopic beds in the Western Pacific. Their close proximity to the Brunhes–Matuyama boundary, with Potaka Pumice of *normal* polarity and Rewa Pumice and Lower Te Muna Tephra of reversed polarity, makes them useful for absolute age control and correlation. The distribution of Rewa Pumice, which we assign to glacial oxygen isotope stage 20, allows direct correlation between a diverse range of sedimentary facies including estuarine/shoreface, fluvial, lacustrine, ox-bow lake, and loess. Its distinctive glass chemistry (FeO c. 2.00 wt%; TiO_2 0.14–0.18 wt %) makes it easily recognisable even when mixed with other tuffs by sedimentary processes. Recognition of the Rewa Pumice close to a reversal boundary allows identification of the Brunhes–Matuyama transition.

The difficulty of obtaining good paleomagnetic data for the Rangitikei section where part of the sequence is overprinted by strong viscous fields (Fig. 4) emphasises the importance of tuff correlation to other localities, such as Huangarua and Mangatewaiiti, where good data have been obtained. Complementary to this, the use of paleomagnetism in tuff correlation studies avoids the miscorrelation of different-aged tuffs of similar glass chemistry, such as demonstrated by Potaka Pumice and Lower Te Muna Tephra in this study.

Glass chemistry and paleomagnetism of tuffs suggest no simple tuff correlation can be made between the Rangitikei section with radiometrically dated tuffs and the stratotype section of the Castlecliffian on the Wanganui coast, and that lithostratigraphic units such as the Kaimatira Pumice Sand cannot be used for correlation.

In summary, the following important correlations have been established: (1) Potaka Pumice to tuff S099 (Mangatewaiiti) and tuff C (in cores RC12–215, RC9–113); (2) Rewa Pumice to Upper Te Muna Tephra (Huangarua) and tuff S175/100 (Mangatewaiiti); and (3) Lower Te Muna Tephra to tuff B (in cores RC12–215, RC9–113) and possibly Kaimatira Pumice Sand (Wanganui coast).

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APPENDIX 1

Sample localities for tuffs analysed in this study. Grid references are those of the metric NZMS 260 map series.

Rangitikei section and Wanganui Basin

Rangitikei tuffs are from the Cheltenham–Hunterville Road.
Pakihikura Pumice, T22/362356.
Mangapipi Ash, T22/355344.
Rewa Pumice, T22/348317.
Potaka Pumice, T22/357311.
Kaukatea Ash, N138/710892, Kaukatea valley.
Kaimatira Pumice Sand, N137/418930, Wanganui coast.

Huangarua section and Wairarapa

Upper and Lower Te Muna Tephras, S27/189895, Huangarua River, Te Muna.
Blue Rock Tuff, S27/128847, Blue Rock Stream, tributary to Ruakokoputuna River.
Popes Head Tephras and Akupe Tephra, S27/244920, Popes Head, Hinakura Road.
Te Whanga Tuff and S020, T26/381157, Te Whanga, SE Masterton.

Mangatewaiiti Stream, Hawke's Bay

S075, U23/808174, Mangatewaiiti Stream Bridge, State Highway 2.
S099, U23/811173, Mangatewaiiti Stream.
S100, U23/816167, Mangatewaiiti Stream.

Western Pacific deep-sea cores

RC9–113, lat. 42° 51′, long. 172° 07′.
RC12–215, lat. 35° 28′, long. 167° 53.5′.